



**Progressive Education Society's Modern College of Engineering, Pune-05. (An Autonomous Institute
Affiliated to Savitribai Phule Pune University) Department of S.Y. B. Tech [AIDS]**

Assignment No: 10

SUBJECT: Java Programming

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Class: SY BTech (AIDS)

Section: A

Roll Number: 26040

Title / Problem Statement :

Develop a JavaFX application that provides a graphical user interface to perform basic integer arithmetic operations. the interface should include: a. two input fields labeled 'number 1' and 'number 2'. b. four buttons labeled '+', '-', '*', and '/'. c. a display field labeled 'result'.

Input :`import javafx.application.Application;`

`import javafx.geometry.Insets;`

`import javafx.scene.Scene;`

`import javafx.scene.control.*;`

`import javafx.scene.layout.GridPane;`

`import javafx.stage.Stage;`

`public class Main extends Application {`

`public void start(Stage stage) {`

`TextField num1 = new TextField();`

`TextField num2 = new TextField();`

`TextField result = new TextField();`

`result.setEditable(false);`

`Button add = new Button("+");`

`Button sub = new Button("-");`

`Button mul = new Button("*");`

`Button div = new Button("/");`

`// Event Handling`

`add.setOnAction(e -> {`

`int a = Integer.parseInt(num1.getText());`

`int b = Integer.parseInt(num2.getText());`

`result.setText(String.valueOf(a + b));`

`});`

```
sub.setOnAction(e -> {  
  
    int a = Integer.parseInt(num1.getText());  
  
    int b = Integer.parseInt(num2.getText());  
  
    result.setText(String.valueOf(a - b));  
  
});
```

```
mul.setOnAction(e -> {  
  
    int a = Integer.parseInt(num1.getText());  
  
    int b = Integer.parseInt(num2.getText());  
  
    result.setText(String.valueOf(a * b));  
  
});
```

```
div.setOnAction(e -> {  
  
    try {  
  
        int a = Integer.parseInt(num1.getText());  
  
        int b = Integer.parseInt(num2.getText());  
  
        result.setText(String.valueOf(a / b));  
  
    } catch (ArithmeticException ex) {  
  
        result.setText("Cannot divide by zero");  
  
    }  
  
});
```

```
GridPane grid = new GridPane();  
  
grid.setPadding(new Insets(10));  
  
grid.setVgap(10);  
  
grid.setHgap(10);  
  
  
grid.add(new Label("Number 1"), 0, 0);
```

```
grid.add(num1, 1, 0);
```

```
grid.add(new Label("Number 2"), 0, 1);
```

```
grid.add(num2, 1, 1);
```

```
grid.add(add, 0, 2);
```

```
grid.add(sub, 1, 2);
```

```
grid.add(mul, 0, 3);
```

```
grid.add(div, 1, 3);
```

```
grid.add(new Label("Result"), 0, 4);
```

```
grid.add(result, 1, 4);
```

```
Scene scene = new Scene(grid, 300, 250);
```

```
stage.setTitle("Simple Calculator");
```

```
stage.setScene(scene);
```

```
stage.show();
```

```
}
```

```
public static void main(String[] args) {
```

```
    launch(args);
```

```
}
```

```
}
```

Output :

The image shows an IDE window with the following components:

- EXPLORER:** Shows the project structure with 'ASSIGNMENT_10' containing 'Main.class' and 'Main.java'.
- EDITOR:** Displays the code for 'Main.java'. The code imports `javafx.application.Application`, `javafx.geometry.Insets`, `javafx.scene.Scene`, `javafx.scene.control.*`, `javafx.scene.layout.GridPane`, and `javafx.stage.Stage`. It defines a `Main` class extending `Application` with a `start(Stage stage)` method. Inside the method, it creates `TextField` objects for numbers and results, and `Button` objects for addition, subtraction, multiplication, and division. It also includes event handling logic to parse the input and calculate the result.
- Simple Calculator Window:** A small window titled 'Simple Calculator' with two input fields for 'Number 1' (10) and 'Number 2' (5), buttons for '+', '-', '*', and '/', and a 'Result' field showing '15'.
- TERMINAL:** Shows the command to compile and run the application:

```
PS D:\College\College Coding\Java\assignment_10> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_10> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javaafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
```

The image shows an IDE window with the following components:

- EXPLORER:** Shows the project structure with 'ASSIGNMENT_10' and 'Main.java'.
- EDITOR:** Displays the code for 'Main.java'. The code imports JavaFX classes and implements a simple calculator with two text fields for numbers, buttons for addition, subtraction, multiplication, and division, and a result text field.
- Simple Calculator Window:** A small window titled 'Simple Calculator' with two input fields labeled 'Number 1' and 'Number 2', buttons for '+', '-', '*', and '/', and a 'Result' field showing '50'.
- PROBLEMS:** Shows warnings related to the JavaFX module system.
- Terminal:** Displays the commands used to compile and run the application.

```
1 import javafx.application.Application;
2 import javafx.geometry.Insets;
3 import javafx.scene.Scene;
4 import javafx.scene.control.*;
5 import javafx.scene.layout.GridPane;
6 import javafx.stage.Stage;
7
8 public class Main extends Application {
9
10     public void start(Stage stage) {
11
12         TextField num1 = new TextField();
13         TextField num2 = new TextField();
14         TextField result = new TextField();
15         result.setEditable(false);
16
17         Button add = new Button("+");
18         Button sub = new Button("-");
19         Button mul = new Button("*");
20         Button div = new Button("/");
21
22         // Event Handling
23         add.setOnAction(e -> {
24             int a = Integer.parseInt(num1.getText());
25             int b = Integer.parseInt(num2.getText());
```

PS D:\College\College Coding\Java\assignment_10> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java

PS D:\College\College Coding\Java\assignment_10> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main

WARNING: A restricted method in java.lang.System has been called

WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)

WARNING: Use --enable-native-access=java:fx.graphics to avoid a warning for callers in this module

WARNING: Restricted methods will be blocked in a future release unless native access is enabled